

CHEMISTRY 101 · RECITATION HOMEWORK

Homework 1: Atoms & Air

Name: _____

Date: _____

Instructions: About 20 minutes. You may look back at your Class Notes sheet. Part A is counting practice — go slowly and check each row. Total: **23 points**.

Part A — Count the atoms

1. For each molecule, count the atoms of each element, then the total. (Remember: the little number counts the atom right before it; no number means 1. Write 0 if an element isn't there.)

(6 pts)

Formula	C	H	O	N	Total
H ₂ O (water)					
CO ₂ (carbon dioxide)					
O ₂ (oxygen)					
N ₂ (nitrogen)					
NH ₃ (ammonia)					
C ₆ H ₁₂ O ₆ (glucose — sugar)					

Part B — Draw a molecule

2. Draw CO₂ and H₂O as ball diagrams, like the ones in class. Use the legend: oxygen = red (or label "O"), carbon = black ("C"), hydrogen = light blue ("H"). Label each molecule with its formula. (2 pts)

Part C — The air, in percent

3. Out of every 100 particles of air, about how many are nitrogen? About how many are oxygen? (2 pts)

4. Imagine a room with 100 balloons of air. If 21 of them are oxygen, how many balloons are NOT oxygen? (1 pt)

5. A NICU baby is “on 40%.” Is that MORE or LESS oxygen than room air? How do you know? (2 pts)

Part D — Which way does it go?

6. In the lungs, mark what happens to each gas with an X: goes INTO the blood, goes OUT of the blood into the air, or is mostly ignored (in and right back out). (3 pts)

Gas	INTO the blood	OUT of the blood	Mostly ignored
Oxygen (O ₂)			
Carbon dioxide (CO ₂)			
Nitrogen (N ₂)			

7. Why must the body get rid of carbon dioxide? (1 pt)

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- A. Dissolved in blood it forms an acid, and too much acid disrupts the body's chemistry.
- B. It takes up room in the blood that is needed for carrying sugar to the cells.
- C. It is a heavier gas than oxygen, so it would settle in the legs and feet.
- D. It blocks the red and infrared light that the pulse oximeter needs to read.

Part E — Flashback to NICU Tutor

8. Name the NICU machine that measures oxygen saturation using light. (1 pt)

9. What is surfactant, and why do premature babies need it as a medicine? (2 pts)

Part F — Think like a doctor

10. A preemie's blood gas test shows too much carbon dioxide in the blood. (a) What is this baby probably NOT doing well enough? (b) Name one NICU machine that could help, and say in one sentence what it does. (3 pts)

Score: _____ / 23 points

Professor's comment:
